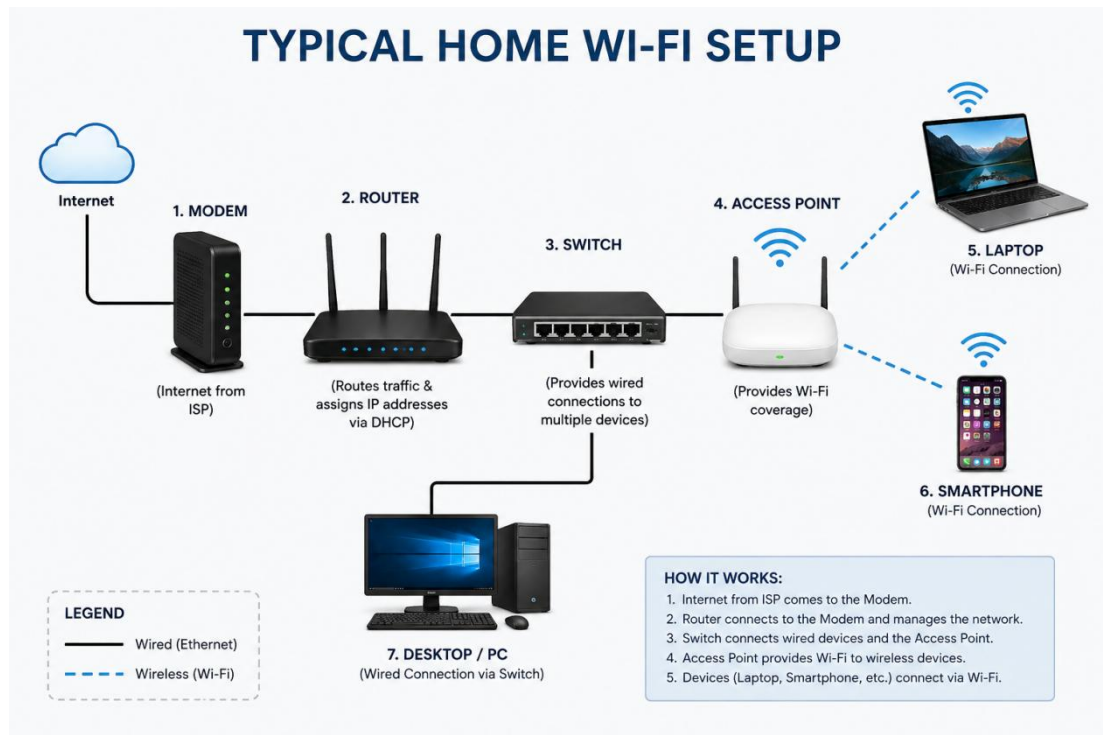


MODULE 3

ASSIGNMENT 3.1

01 . Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

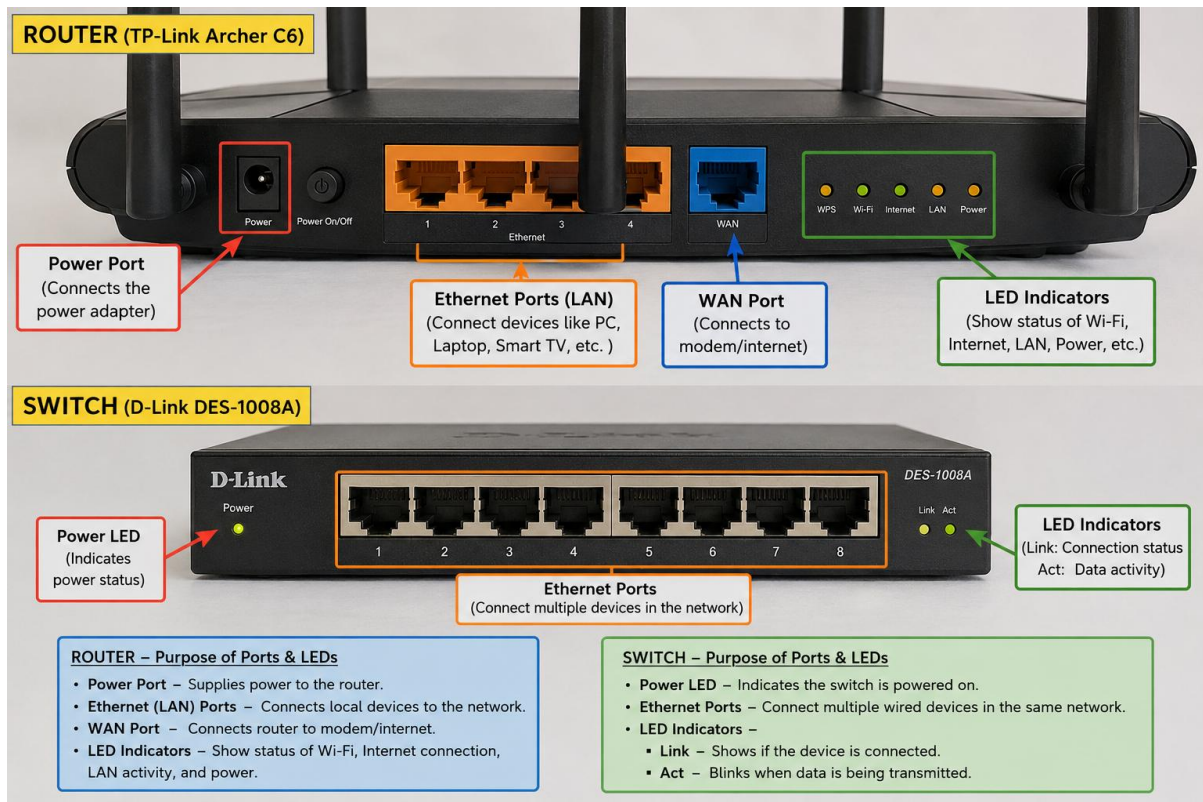


-- In a normal home network, the internet first comes from the ISP (Internet Service Provider) into the modem. The modem changes the signal so that our devices can understand it and sends it to the router. The router shares this internet with all devices and also creates the home Wi-Fi. A switch can be added to the router to connect more wired devices, like a desktop PC. An access point is connected to the router to give stronger Wi-Fi coverage in bigger houses, and devices like laptop and smartphone connect to it wirelessly.

02. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Device	Main Function	Real-Life Example
Modem	Converts the internet signal coming from the ISP (through telephone/cable line) into a digital signal that our home devices and router can use, and the other way around.	The small box given by Jio/Airtel/BSNL that we connect to the wall socket for internet.
Router	Takes the internet from the modem and distributes it to all devices in the house, and creates the Wi-Fi network so devices can connect wirelessly.	At home, to share one internet connection with mobile, laptop and TV together.
Switch	Connects many wired devices in the same network and sends data only to the correct device instead of sending it to everyone.	In an office or computer lab to connect many computers with LAN cables.
Access Point	Gives wireless (Wi-Fi) access to devices and increases the Wi-Fi coverage area of a network.	In a big office, school or mall, where one router's Wi-Fi cannot reach every corner.
Network Cable	Physically carries the data between two networking devices, like from computer to router or router to switch.	The blue LAN cable used to connect a desktop computer to a router at a cyber cafe.

03 Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



04. Imagine you are setting up a small gaming cafe network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Scenario: Gaming Cafe Network

I am setting up a small gaming café for **IPL streaming and multiplayer gaming**. I would connect the **router first to the ISP line** because the router provides Internet access, assigns IP addresses, and manages traffic between the cafe network and the Internet.

Then, I would connect the router to an **8-port or 16-port network switch** using an Ethernet cable. The **8 gaming PCs** would each be connected to separate ports on the switch using Ethernet cables.

Network layout:

ISP → Router → Switch → 8 Gaming PCs

The **router** is needed to connect the local cafe network to the Internet, while the **switch** allows all 8 PCs to communicate efficiently on the local network. A wired switch connection is preferred for gaming because it provides a more stable and lower-latency connection than relying on Wi-Fi.